

HORTA, Portugal

WMO: 085060

Lat: 38.5297N

Lon: 28.6286W

Elev: 62

StdP: 100.58

Time Zone: -1.00 (AZR)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.8	10.7	2.8	4.6	10.8	4.3	5.2	11.6	16.4	15.3	14.9	15.2	5.4	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	26.1	22.0	25.4	21.6	24.7	21.2	23.1	24.8	22.6	24.4	22.2	23.9	3.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.3	24.1	22.0	16.8	23.8	21.6	16.3	23.4	68.5	25.0	66.8	24.4	65.0	23.9	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	12.0	10.5	7.9	27.9	1.2	0.9	7.0	28.5	6.3	29.0	5.6	29.5	4.7	30.2		
(5)			5.8	24.0	0.9	0.5	5.2	24.4	4.7	24.7	4.2	25.0	3.6	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	15.0	14.6	14.7	15.2	16.9	19.1	21.5	22.8	21.7	19.4	17.2	15.5	(6)
(7)		DBStd	3.26	1.47	1.52	1.46	1.41	1.31	1.54	1.28	1.12	1.51	1.91	1.88	1.87	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	614	103	106	112	93	48	9	0	0	1	11	44	88	(9)
(10)		CDD10.0	2854	156	128	147	157	212	273	358	397	351	291	214	171	(10)
(11)		CDD18.3	428	0	0	0	0	3	33	99	139	101	43	8	1	(11)
(12)		CDH23.3	597	0	0	0	0	0	11	113	327	132	13	0	0	(12)
(13)		CDH26.7	7	0	0	0	0	0	0	1	5	1	0	0	0	(13)
(14)	Wind	WSAvg	4.8	5.4	5.3	5.6	5.1	4.6	4.2	3.7	3.6	4.1	5.1	4.9	5.7	(14)
(15)	Precipitation	PrecAvg	980	109	95	91	65	56	49	34	53	87	116	113	117	(15)
(16)		PrecMax	1449	247	174	223	150	152	121	68	207	248	290	226	277	(16)
(17)		PrecMin	778	16	21	14	8	1	11	1	3	17	22	8	22	(17)
(18)		PrecStd	139	53	41	53	38	36	29	18	37	60	55	58	55	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	18.5	18.7	19.9	21.4	24.5	26.2	27.2	26.5	24.6	22.2	19.9	(19)
(20)			MCWB	15.9	15.9	16.2	16.0	18.3	21.1	22.0	22.5	22.5	20.8	19.5	18.0	(20)
(21)		2%	DB	18.0	17.6	17.8	18.9	20.4	23.2	25.2	26.3	25.5	23.5	21.1	19.1	(21)
(22)			MCWB	16.1	15.4	15.5	15.8	17.6	20.0	21.4	22.1	21.9	20.4	18.5	17.4	(22)
(23)		5%	DB	17.5	17.1	17.2	18.1	19.7	22.3	24.5	25.7	24.8	22.7	20.2	18.6	(23)
(24)			MCWB	15.8	15.2	15.2	15.3	17.1	19.2	21.0	21.8	21.3	20.0	18.0	16.9	(24)
(25)		10%	DB	17.1	16.7	16.7	17.4	19.1	21.5	23.8	25.1	24.0	22.0	19.7	18.1	(25)
(26)			MCWB	15.6	14.9	14.9	15.0	16.7	18.8	20.6	21.5	20.8	19.7	17.6	16.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	16.8	17.0	17.2	19.2	22.0	23.2	23.9	23.5	22.0	20.4	18.7	(27)
(28)			MCDWB	18.0	17.4	18.0	18.4	20.4	24.1	24.9	25.5	25.4	23.3	21.4	19.3	(28)
(29)		2%	WB	16.8	16.3	16.5	16.7	18.5	21.1	22.6	23.2	22.8	21.3	19.2	18.1	(29)
(30)			MCDWB	17.4	17.0	17.3	17.7	19.7	22.3	24.2	24.9	24.5	22.5	20.2	18.8	(30)
(31)		5%	WB	16.4	15.9	16.1	16.3	18.0	20.2	22.0	22.9	22.3	20.8	18.7	17.6	(31)
(32)			MCDWB	17.2	16.7	16.8	17.4	19.2	21.5	23.5	24.6	24.0	22.1	19.8	18.4	(32)
(33)		10%	WB	16.0	15.5	15.7	15.8	17.4	19.6	21.5	22.4	21.7	20.3	18.2	17.0	(33)
(34)			MCDWB	16.9	16.4	16.5	17.0	18.6	20.9	23.0	24.2	23.4	21.6	19.5	17.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.5	3.7	3.7	4.2	4.2	4.5	4.7	4.9	4.5	3.9	3.4	3.6	(35)
(36)			MCDBR	3.9	3.7	3.7	4.5	4.6	5.2	5.2	5.4	5.0	4.2	3.6	3.6	(36)
(37)			MCWBR	3.4	3.2	3.1	3.3	3.2	3.4	3.1	3.1	3.1	3.0	2.8	3.3	(37)
(38)		5% WB	MCDWB	3.0	3.5	3.2	3.9	4.0	4.5	4.4	4.7	4.3	3.7	3.2	3.2	(38)
(39)			MCWBR	3.3	3.5	3.0	3.2	3.1	3.2	3.2	3.2	3.2	3.2	3.0	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.354	0.361	0.391	0.384	0.379	0.378	0.373	0.368	0.366	0.368	0.354	0.355	(40)	
(41)		taud	2.506	2.486	2.388	2.432	2.475	2.511	2.536	2.573	2.579	2.554	2.539	2.482	(41)	
(42)		Ebn at Noon	819	863	868	895	902	899	900	900	884	846	813	787	(42)	
(43)		Edh at Noon	80	91	111	112	109	106	102	96	91	85	76	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.01	2.85	3.96	5.16	5.94	6.33	6.52	6.15	4.82	3.35	2.26	1.72	(44)	
(45)		RadStd	0.16	0.15	0.28	0.46	0.46	0.42	0.36	0.25	0.24	0.21	0.13	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	+1.02	N/A	+0.35	+0.52	N/A	N/A	N/A	N/A

Nomenclature: See separate page